

cTVF Project Score Synthesis: Replies to Questions and Comments

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Global

Q: BENCHMARKS FOR ALL SUPPLY CHAINS?

Yes. The cTVF constants (pi, phi, sqrt(2), ln(2), e) are universal mathematical constants and their convergence values apply identically across all supply chains. The benchmark thresholds are:

- $\phi^{-1} = 61.8\%$ is the sustainability floor for all chains (D:C ratio = phi).
- $1 - \phi^{-1} = 38.2\%$ is the breakdown threshold for all chains (D:C ratio = ϕ^2).
- The operational axis uses $\pi/4 = 78.5\%$ as the reference for phase-level integration.

These are not supply-chain-specific calibrations. They derive from the Tholonic N-D-C framework's mathematical structure, making them valid benchmarks across all supply chains without modification. A chain-specific narrative explains what each axis measures *in that context* (e.g., the **e** axis is financial coupling for chocolate, trade structure for olive oil), but the threshold values do not change.

Cocoa Netherlands

Dashboard: https://tvf.tholonia.com/frontend/project/cocoa_netherlands/supply_chain/dashboard.html

Q: Why is the Phase Balance Pi not Phi?

The Phase Balance axis is anchored to **$\pi/4$ (approximately 78.54%)** rather than ϕ^{-1} (61.8%) because it measures *operational integration*, not proportional sustainability. $\pi/4$ is the natural convergence for a balanced D-C oscillation across a phase boundary when new information is injected at each transition. In Tholonic terms, pi governs the rotational/cyclical integration of constraint and definition across operational steps. Phi governs the *proportional scaling* between levels. They measure orthogonal properties. Using phi for the Phase Balance axis would conflate structural sustainability with operational integration capacity.

Q: What are the upper and lower limits of the other constants, how do they relate to their convergence values, and what does it mean?

For each axis, the natural convergence value *is* the reference:

Axis	Constant	Reference value	% equivalent	Meaning at that value
Operational	$\pi/4$	0.7854	78.5%	Phases are perfectly integrated, constraints match integration capacity

Axis	Constant	Reference value	% equivalent	Meaning at that value
Proportional	1/phi	0.6180	61.8%	Value escalates at the natural golden ratio between phases
Structural	1/sqrt(2)	0.7071	70.7%	Transfer overhead equals the irreducible minimum cost of crossing a phase boundary
Growth	ln(2) / 1	0.6931	69.3%	Transformation phases double value at the natural rate
Financial	1/e	0.3679	36.8%	Or equivalently e^{-1} ; financial coupling at natural decay rate

Scores above the reference indicate the chain is performing better than the natural baseline on that dimension. Scores below indicate drag, extraction, or decoupling relative to the natural optimum. Upper bound is 100% (perfect); lower is 0% (complete breakdown or unmeasured).

Comment: Do not abbreviate "H" to "High" in the Data column

Agreed. The label "H High" in the Data quality field is redundant. Change it to simply the data quality descriptor (e.g., "High") or the actual source type. Abbreviating "H" alongside the word "High" adds noise without meaning.

SN: Growth Value transformation capture (Cocoa Netherlands)

Noted as a supply note for the dashboard. CBE (Cocoa Butter Equivalent) manufacturing creates a distinct value layer that the current phase structure should capture explicitly. IOI's single-actor concentration in grinding limits competitive upward pressure on the growth axis. This structural feature explains why the ln(2) growth score is above the global cocoa average despite the certification premium concentration.

SN: Financial coupling stronger than global (58.3) for Cocoa Netherlands

The Dutch Cocoa Agreement's Sustainability-Linked Bond (SLB) structure and EUDR compliance premiums create traceable financial signals that directly couple supply chain performance to pricing. This lifts the e score relative to the global cocoa benchmark. The NL-specific financial architecture is materially better integrated than the global average, though still in the failure zone.

Q: Phase Balance analysis uses pi/4, explain this and why

Phase Balance is a measure of how well each phase's output integrates into the next phase's input, i.e., the operational handoff quality. Pi/4 is the natural reference because a complete oscillation across a D-C boundary completes in pi/4 radians of the tholonic cycle. When a phase scores at or above 78.5%, the handoff is clean: definition and constraint are in balance at the transfer point. Below 78.5%, the phase is either over-constrained (bottleneck) or under-defined (leakage). This is distinct from the phi-based proportionality check, which measures whether *value itself* scales correctly, not whether the operational handoff is clean.

Action Required: Remove sustainability rating from Phase Balance Analysis

Confirmed action item. The Phase Balance Analysis chart/table must not display a sustainability rating. Sustainability scoring (Coherent / Failure / Breakdown) belongs exclusively to the Phi-axis analysis. Phase Balance scores are operational integration measures and their meaning is "well integrated" or "over/under constrained," not "sustainable" or "unsustainable." Remove the sustainability zone labels from all Phase Balance visualisations across all projects.

SN: **e** axis interpretation differs by project context

Correct. The **e** (Abstract/Financial) axis measures abstract coupling to physical performance, but the *type* of abstraction is project-specific:

- **Chocolate/Cocoa:** **e** = financial coupling (pricing, certifications, financial instruments vs. physical chain performance).
- **Olive oil:** **e** = trade structure coupling (export contracts, provenance certification, PDO/PGI vs. physical flow).

The threshold (61.8%) and zone labels remain the same. The axis label and narrative description should reflect the project-specific interpretation.

Q: Tony's Chocolonely benchmark. What is the potential market share for (1) Living Wage chocolate and (2) High-end quality chocolate?

Tony's Chocolonely is a meaningful independent benchmark because it explicitly prices in living wages for farmers. Estimated addressable markets:

1. **Living Wage chocolate:** The premium/ethical segment represents approximately 3-5% of global chocolate volume but 8-12% of value in mature markets (Netherlands, Germany, UK). Tony's holds roughly 25% of the Dutch ethical chocolate segment. Scaling to Europe, a conservative Living Wage-compliant supply chain could address 5-8% of the total market by volume within a decade under current regulatory trajectory (EUDR, CSDDD).
2. **High-end quality:** Single-origin and fine/flavour cocoa segments represent approximately 5-7% of global volume. The overlap with Living Wage is growing as premium quality and provenance certification increasingly co-occur. The combined segment (Living Wage + High End) likely represents 6-10% of European market value at current growth rates.

These figures should be sourced and tabulated in the Cocoa Netherlands project context as a benchmark reference node.

Olive Oil Andalusia

Recycling Analysis: https://tvf.tholonia.com/frontend/project/olive_oil/supply_chain/recycling_analysis.html

Comment: TVCPI description on the recycling analysis page is incomplete

The recycling/added value from waste products (pomace oil, olive mill wastewater valorisation) is only one application of the TVCPI framework. The full TVCPI definition includes: living wage measurement, provenance integrity, custody transparency, and the full N-D-C balance across phases. Use the definition from the TVA website front page as the canonical description on all project pages. The recycling analysis page should frame itself as "TVCPI applied to waste valorisation" rather than implying that recycling *is* what the TVCPI does.

Q: Explain Holon and B-Chain. What is the impact on R-P?

A **Holon** is any entity that is simultaneously a whole (complete in itself) and a part of a larger system. In the supply chain context, each phase is a holon: it has internal N-D-C balance (its own Negotiation, Definition, Contribution dynamic) and simultaneously participates in the chain's overall balance. The **B-Chain** (Balance Chain) is the sequence of holon-level balance scores across phases, which collectively determine the chain-average cTVF score.

Impact on R-P (Resource-to-Product ratio): When a phase-level holon is unbalanced (score below 61.8%), it introduces friction at the R-P boundary. Specifically:

- An over-constrained holon ($D \gg C$) reduces throughput, raising effective R-P cost.
- An under-defined holon ($C \gg D$) introduces leakage, reducing yield at R-P conversion.
The chain's structural integrity ($\sqrt{2}$ axis) directly reflects the aggregate R-P efficiency across all transfer holons.

Action Required: Add climate factors as production bottleneck

Confirmed action item. The Olive Oil Andalusia supply chain model must include climate stress (drought frequency, temperature extremes, frost) as an explicit constraint at Phase 1 (Primary Production). Andalusia has experienced severe drought-driven harvest reductions in 2022-2023 (approximately 50% below historical average). This must appear as:

- A bottleneck annotation on the production phase.
- A climate risk metric in the schema (e.g., `climate_constraint_index`, `drought_impact_%`).
- A note in the Phase Balance analysis explaining the structural depression of production-phase scores.

Action Required: Include high-end benchmark (Las Valdesas, Andalusia)

Confirmed action item. Las Valdesas (<https://www.aceitedelasvaldesas.com/>) represents an independently operated, direct-from-bodega, single-estate premium olive oil benchmark. Key differentiators noted:

- Direct-to-consumer shipping in 10 days, packaged only in cardboard (no foam, no plastic insert).
- No intermediary distributor step, which compresses the supply chain by 1-2 phases.
- Provenance fully traceable to single estate.

Add as a benchmark node in the Olive Oil Andalusia project, analogous to Tony's Choclonely in the cocoa projects. Map its phase structure and compare against the standard Andalusian supply chain cTVF scores to quantify the value of chain compression and direct trade.

Cocoa International

Recycling Analysis: https://tvf.tholonia.com/frontend/project/cocoa_international/supply_chain/recycling_analysis.html

SN: TVPCI Primary Chain summary (Cocoa International, Balance 89.2%)

Noted. The primary chain (farm to retail) scores 89.2% operational balance, with the binding constraint in institutional, regulatory, and custody gaps rather than physical logistics. The high operational score reflects that the physical movement of cocoa is well-optimised; the failures are in information asymmetry, certification opacity, and custody handoff documentation.

SN: e-layer score 58.3 (failure zone) explanation (Cocoa International)

The **e** score of 58.3 falls below the 61.8% sustainability floor. This indicates that sustainability certifications (Rainforest Alliance, UTZ, Fairtrade) trade as financial instruments that are only weakly coupled to verifiable on-farm performance. Premium price signals do not reliably flow back to producers; certification bodies act as intermediaries that absorb much of the coupling signal. Until certification audit trails are directly linked to payment flows (e.g., blockchain-anchored, or SLB-style performance contracts), this axis will remain in the failure zone for international cocoa.

Shea

Action Required: Improve communication of the Phase Balance radar chart

Confirmed action item. The current chart description is technically correct but too dense. Proposed clearer framing for the reading guide:

"Each ring on this chart marks a cTVF threshold derived from phi (the golden ratio). The outer ring at 61.8% is the sustainability floor: phases scoring above this value are in the coherent zone (value and constraint are in natural proportion). The inner ring at 38.2% is the breakdown threshold: phases scoring below this are structurally broken. The region between 38.2% and 61.8% is the failure zone: constrained but not yet collapsed."

Remove the table-format zone key and replace with the inline annotation approach. Polygon edge colours should remain; the ring labels need only say "61.8% Sustainability Floor" and "38.2% Breakdown Threshold."

Q: Can we check the maths of phi relative to percent points? (Zone boundaries)

The derivation is:

- $\Phi = (1 + \sqrt{5}) / 2 = 1.6180...$
- $1/\Phi = \Phi - 1 = 0.6180... = \mathbf{61.8\%}$ (sustainability floor: D:C = Φ , i.e., definition is Φ times contribution)

- $1 - 1/\phi = 1 - 0.618 = 0.382 = \mathbf{38.2\%}$ (breakdown threshold: $D:C = \phi^2 = 2.618$)

The percentage points are exact (to 3 significant figures) because ϕ 's reciprocal is ϕ minus 1. There is no rounding approximation: 61.8% and 38.2% sum exactly to 100% and reflect the self-similar nature of the golden ratio. The maths is correct.

On the coherent zone description: "The greater the percentage above 61.8%, the closer to perfect D:C balance" is correct directionally. Perfect balance would be $D:C = 1$, corresponding to 50%. The ϕ floor at 61.8% is not "perfect" balance, it is the *minimum sustainable* asymmetry. The coherent zone (61.8-100%) contains a range from minimum sustainability up to hypothetical perfect operational efficiency. This nuance should be explained in the chart legend.

Q: Is the failure zone (38.2-61.8%) "over-constrained"?

Partially. The failure zone indicates that $D:C$ has exceeded ϕ (1.618), meaning Definition is more than ϕ times Contribution. In supply chain terms, this manifests as over-extraction, under-investment, or both. "Over-constrained" is the correct description for the $D:C$ imbalance direction (definition/extraction dominating contribution/regeneration). The breakdown zone (below 38.2%) represents $D:C$ exceeding ϕ^2 (2.618), where the imbalance is so severe that the system cannot self-correct.

cTVF Axis Interpretations for Shea

The spreadsheet contains expanded interpretations for each Shea axis. Confirming these are accurate and should be used as the basis for dashboard annotations:

- **ϕ (Proportional 53.6):** In the failure zone. Phase transitions are disproportionate: value is not escalating at the natural golden ratio between phases. Priority: identify which phase transitions show margin collapse (likely Phase 2-3, primary processing to intermediary).
- **$\sqrt{2}$ (Structural 38.1):** At the breakdown threshold. Transfer phases are functioning as extraction points. The overhead at each boundary far exceeds the natural $\sqrt{2}$ cost. This is the highest-priority structural intervention point.
- **$\ln(2)$ (Growth 39.4):** Near the breakdown threshold. Processing and transformation phases have been converted from growth engines into rent-extraction bottlenecks. Artificial barriers (broker concentration, lack of direct trade access) are damping the natural value-doubling dynamic.
- **e (Abstract 54.0):** In the failure zone. Some financial instruments or certifications are outrunning the physical chain. Monitor for certification gap widening (certifications issued against smallholder farms without verified audit trails).
- **Primary risk:** The 43.5 percentage-point gap between Operational (81.6%) and Structural (38.1%) defines the chain's key vulnerability. Structural improvements (reducing transfer-phase extraction, improving custody documentation) will have the highest system-wide impact.

Q: Can we work toward questioning these points for more clarification?

Yes. The approach should be: for each axis score, state (1) what the score means in physical supply chain terms, (2) which specific phases or actors are responsible for the deviation, and (3) what intervention would move the score toward the reference value. The current axis narratives (rows 41-49) are a good start but are written at the system level. Phase-level attribution is the next analytical step.

Gold

SN: Layer selection description

The current description ("Choose a layer: Supply Chain models physical flow...") is accurate but the framing "Choose a layer" is a UI instruction, not a conceptual description. The description should explain *why* the layers are separated, not just that they exist. Suggested revision: "The Supply Chain, System Lifecycle, and Value Chain analyses are kept strictly separate because they measure different properties: physical custody and constraints, maintenance and closure obligations, and economic margins respectively. Mixing these layers obscures accountability."

Comment: TVPCI description is incomplete and confusing (Gold)

As with Olive Oil: the current "Primary custody chain / The bounding, defining forward structure / What was extracted, claimed, and commercialised" description is accurate but covers only the custody dimension of the TVCPI. Use the full definition from the TVA website front page. The TVCPI encompasses living wage measurement, provenance integrity, financial coupling, and environmental accounting in addition to custody transparency.

Comment: Holon structure referenced without explanation (Gold)

Every reference to "holon" or "B-chain" in the Gold analysis must include an inline definition or a link to the framework glossary. See the Olive Oil section above for the canonical definition. This applies to all projects: do not use Tholonic framework terminology without a first-use explanation on the same page.

Q: What do ADI and True Value % margins mean in the Financial Report? Why does the True Value margin decrease when the commodity price (C) increases?

ADI (Adjusted Distribution Index): The ADI measures how equitably value is distributed across chain participants relative to their contribution. An ADI of 1.0 means each actor receives value proportional to their N-D-C contribution. ADI > 1 means downstream actors are capturing disproportionate value; ADI < 1 means upstream actors (typically producers) are under-compensated.

True Value margin: The True Value margin includes internalised externalities (environmental cost, living wage gap, social cost) that are excluded from the conventional market margin. It is calculated as:

$$\text{True Value margin} = \text{Conventional margin} - (\text{Environmental cost} + \text{Social cost gap}) / \text{Revenue}$$

Why True Value margin decreases as commodity price rises: When the commodity price (C price, e.g., gold spot price) rises, conventional margins appear to improve for midstream and downstream actors. However, the social and environmental externalities embedded in extraction (water use, land rehabilitation obligation, artisanal miner displacement) scale with extraction volume, not price. If the price rise is not accompanied by increased investment in externality mitigation, the True Value margin shrinks even as the conventional margin grows. This is the financial decoupling signal: the **e** axis score of 37.0 for Gold reflects precisely this dynamic.

cTVF Axis Interpretations for Gold

Confirming the axis narratives are accurate:

- **pi (Operational 82.5):** Strong. Phase-level D-C balance is well matched. The operational extraction-to-refining chain is efficient.
- **phi (Proportional 49.5):** Failure zone. Value escalation between phases is disproportionate. The gap between mine-gate value and LBMA price reflects extraction at the proportional boundary.
- **sqrt(2) (Structural 72.1):** Coherent. Custody transfer and logistics are largely symmetric. This is the strongest non-operational dimension.
- **ln(2) (Growth 40.4):** Near failure threshold. Transformation phases are not doubling value at the natural rate. Cost, certification burden, and buyer power are damping the growth dynamic.
- **e (Abstract 37.0):** Breakdown zone. Paper claims (unallocated accounts, ETF holdings, derivatives) significantly exceed what the physical chain supports. This is the most dangerous failure mode in the Gold chain and the primary systemic risk.

Photosynthesis

Comment: Chart zones use 40%/70% thresholds, should use 61.8%/38.2%

Confirmed action item. The Photosynthesis radar chart uses 40% and 70% zone boundaries, which are arbitrary rounded approximations. Replace with the phi-derived boundaries used in all other analyses: **61.8%** (sustainability floor) and **38.2%** (breakdown threshold). The note "outer zone (70-100%) is coherent; middle zone (40-70%) is stressed; inner zone (<40%) is failure" must be updated to match the standard cTVF zone definitions.

Comment: Same correction applies to AUBEB

Confirmed action item. The AUBEB analysis also uses the 40%/70% approximate thresholds. Apply the same correction: replace with 61.8% and 38.2% phi-derived boundaries and update all zone labels to Coherent / Failure / Breakdown with the standard cTVF definitions.

Note: The Photosynthesis chart correctly notes that **e** (Abstract) = 0 means *unmeasured*, not failed. This distinction is important and should be retained and applied consistently across any chain where an axis is genuinely inapplicable (as opposed to scoring zero due to absence of performance).

Summary of Action Items

#	Action	Applies to
1	Remove sustainability rating from Phase Balance Analysis	All projects
2	Replace "H High" data label with full descriptor	Cocoa Netherlands
3	Use full TVCPI definition from TVA website front page	Olive Oil, Gold, all projects

#	Action	Applies to
4	Add climate factors (drought) as production bottleneck	Olive Oil Andalusia
5	Add Las Valdesas as high-end benchmark	Olive Oil Andalusia
6	Add Tony's Chocolonely market share benchmark data	Cocoa Netherlands
7	Add inline holon/B-chain explanation on all pages that use the term	All projects
8	Replace 40%/70% zone thresholds with 61.8%/38.2%	Photosynthesis, AUBEB
9	Explain ADI and True Value margin in the Gold Financial Report	Gold
10	Add phase-level attribution to cTVF axis narratives (not just chain-level)	All projects
11	Clarify that e axis context differs by project (financial vs. trade structure)	All projects
12	Add upper/lower limit reference table for all five constants	All projects